

Muhammed Nabhan

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Portfolio | LinkedIn

SUMMARY

B.Tech graduate in Artificial Intelligence and Machine Learning with hands-on experience developing AI/ML, computer vision, NLP, RAG, chatbot, and automation solutions across three internships. Experienced with Python, TensorFlow, Vision Transformers, explainable AI, LLM fine-tuning, RAG pipelines, FastAPI, Flask, Docker, and n8n. Co-author of a government-funded KSCST research project on explainable AI for multispectral crop stress detection, published in 2025. Experienced in building end-to-end solutions from prototyping through deployment in collaborative engineering environments.

EDUCATION

Christ University

B.Tech in Artificial Intelligence and Machine Learning

Bengaluru, India

Aug 2022 – May 2026

- **Coursework:** Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Python
- Selected for State-Level Poster Presentation & Exhibition, SPP 2025–26 (Government of Karnataka)

TECHNICAL SKILLS

- **Programming:** Python, HTML, CSS, JavaScript
- **AI/ML:** TensorFlow, Keras, Scikit-learn, NLTK, RAG, Generative AI, LLMs, Vision Transformers
- **Computer Vision:** OpenCV, Pillow, YOLOv8, MediaPipe
- **Explainable AI:** SHAP, Grad-CAM
- **Backend & APIs:** FastAPI, Flask, Gradio, REST APIs
- **AI/Automation:** n8n, OpenClaw, GitHub Copilot, Claude Code, Cursor, Copilot Studio
- **Development & Cloud:** Git, GitHub, Docker, Google Colab, AWS

PROJECTS

Explainable Multispectral Crop Stress Detection

Explainable AI | Computer Vision | Vision Transformers

- Built an explainable AI system using Sentinel-2 satellite imagery and Vision Transformers to detect crop stress before visible symptoms appear.
- Implemented SHAP and Grad-CAM for model interpretability and visual explanation of predictions.
- Implemented a RAG-based chatbot as an additional project feature and co-authored the associated research paper.
- **Tools:** Python, TensorFlow/Keras, Vision Transformers, OpenCV, Gradio, SHAP

Multilingual LLM-Based System for Automated Patient Assistance

Large Language Models | Voice AI | Multilingual AI | Healthcare Automation

- Built a multilingual voice AI system supporting English, Arabic, Hindi, and Malayalam for patient intake, appointment booking, returning-patient assistance, and emergency escalation.
- Developed a state-driven conversational pipeline using Groq LLM, Whisper, Deepgram, and Edge TTS with doctor recommendation, patient profile retrieval, automated email/PDF confirmation, and emergency detection.
- Implemented sentence-level LLM streaming with speculative TTS prefetching for 60–80% lower perceived latency, along with regression testing for hallucination, prompt injection, and multilingual conversation scenarios.
- **Tools:** Python, FastAPI, Groq, Deepgram, Whisper, Edge TTS, LLMs, SSE, Prompt Engineering, ReportLab, SMTP

RAG-Based Cultural Heritage Knowledge Retrieval

RAG | Information Retrieval | Cultural Heritage APIs

- Built a RAG system that retrieves information from multiple cultural heritage APIs and generates context-aware answers grounded in historical sources.
- Implemented multi-source retrieval, data processing, and source attribution for traceable, evidence-backed responses.
- **Tools:** Python, Streamlit, RAG, Pandas, Europeana API, Wikidata SPARQL API, Smithsonian Open Access API

Real-Time Hand Gesture Recognition for HCI

Computer Vision | Gesture Recognition

- Built a real-time gesture-based computer control system for cursor movement, volume control, and click operations using hand gestures.
- Integrated MediaPipe hand tracking with OpenCV for real-time gesture recognition and video processing.
- **Tools:** Python, MediaPipe, OpenCV, NumPy

EXPERIENCE / INTERNSHIPS

AI Engineering Intern – Innobayt Innovative Solutions

Dec 2025 – Feb 2026

AI Engineering

Dubai, UAE

- Developed multiple AI agents and small-scale AI models for business automation workflows.
- Built an agent-based workflow to fetch data and generate automated messages, along with an automated invoice creation bot.
- Worked with Python, GitHub, Copilot Studio, and VS Code to prototype and implement AI-driven automation solutions.

Product Documentation Intern – OHC

Feb 2025 – Aug 2025

Documentation, UI Testing & Validation

Remote

- Created technical documentation and tutorials for the CARE website at OHC Network.
- Performed UI testing to improve usability and validated module outputs through local deployment.
- Worked with GitHub, VS Code, Google Docs, and documentation tools in a collaborative development environment.

Intel® Unnati AI Trainee

May 2024 – Jul 2024

LLM Fine-tuning & GenAI Chatbot Development

Remote

- Developed an AI-powered mental health chatbot using the Intel neural-chat-7b-v3 model and deployed it on Intel Developer Cloud.
- Fine-tuned NLP models for personalized healthcare query responses using a custom dataset.
- Used Python, Intel Neural Chat, Flask, and Google Colab for model development and deployment.

CERTIFICATIONS

- **AWS Cloud Practitioner** – Amazon Web Services
- **Data Science for Engineers** – NPTEL
- **Retrieval-Augmented Generation for Enhanced AI Outputs** – IBM SkillsBuild
- **Computer Vision** – IIT Tirupati
- **Introduction to Cybersecurity** – Cisco Networking Academy

ACHIEVEMENTS

- **Research Publication:** Co-authored “Effective Explainable Model to Detect Multispectral Crop Stress Using ViT” (Published 2025).
- **KSCST Research Project:** Government-funded research project under the 49th SPP, Government of Karnataka (2025–26).
- **State-Level Presentation:** Presented research at the State-Level Poster Presentation & Exhibition, SPP 2025–26.